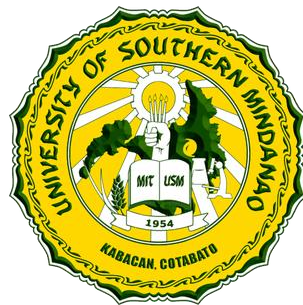


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Customer Segmentation Using K-Means Clustering Based on Income and Spending Behavior



Bachelor of Science in Computer Science  
CS Elec01: Computational Science

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## **INTRODUCTION**

Companies and organizations still collect customer data to discover what customers do to help with making the best decisions possible. One crucial aspect in analyzing the customer behavior is segmentation. Customer segmentation refers to the process of dividing customers into segment(s) or group(s) based on sharing one or more of their characteristics such as their buying patterns, buying behaviors, income levels per year etc. Using segments, the company can detect variety types of customers and treat each customer segment accordingly.

Through data analysis and machine learning technologies, businesses have been enabled to efficiently manage and analyze customer data on a large scale. One of the machine learning algorithms commonly applied in the process of segmentation is the K-Means Clustering algorithm. The K-Means algorithm is an unsupervised machine learning algorithm where data points are clustered according to similarities among them and there are no pre-defined classes or categories within the dataset to be found, which the algorithm has been trained for.

This research is to be carried out using the K-Means Clustering algorithm on data points related to customer annual income and their shopping habits to define and separate different groups of customers based on those variables. In doing so, the researcher will examine the characteristics of each segment of customers and determine their buying patterns. The researcher will make use of a secondary dataset on customer information for the demonstration of how a segmentation could work for the market.

The researcher chose to study this because customer segmentation is used frequently in current business and marketing areas. It also illustrates how machine learning could be utilized in the solving of real-world problems via the manipulation and representation of data.

## **STATEMENT OF THE PROBLEM**

This study focuses on building a customer segmentation model based on the K-Means Clustering Algorithm. More precisely, the following questions are being addressed:

1. How can customer data be clustered using K-Means Clustering Algorithm?
2. What is the trend within customers spending and income behavior?
3. How can business make analysis of groups of customers based on customer segmentation?
4. How efficient is K-Means Clustering in classifying customers into clusters?

## **OBJECTIVES OF THE STUDY**

### **General Objective**

To design a customer segmentation system applying K-Means Clustering Algorithm.

### **Specific Objectives**

1. To gather and process the customers' data regarding annual income and spending score.
2. To cluster the customers using K-Means Clustering Algorithm.
3. To illustrate the clusters of customers by graphs.
4. To observe the customer behavior among the clusters.
5. To assess the efficiency of K-Means Clustering to segment customers.

## **SIGNIFICANCE OF THE STUDY**

This project will be useful to:

- Business & companies

Can learn about customer behavior and target group for marketing strategies.

- Students & Researchers

Can serve as a source for future researchers in the field of machine learning, clustering and customer analysis.

- Future Developers

It will help future developers in creating better and advanced customer analysis systems through machine learning techniques.

- Researchers

This project gives knowledge about K-Means Clustering applied in customer segmenting and analysis.

## **SCOPE AND DELIMITATION**

This research aims to segment customers using K-Means Clustering Algorithm. The customers' annual income and spending score are two attributes which the researchers will use for grouping them into cluster. This research does not cover the customers' transactions online but only consists of customers' data which are being synthesized for educational and demo purposes. There will be no real time online customers' transactions and real time predictions. The main area that the research is focusing on is how to segment customers using clustering algorithms and view their groups in form of graphs and charts.

## RESEARCH METHODOLOGY

### RESEARCH DESIGN

This research uses a descriptive quantitative research design. Descriptive because the aim is to define the customer segments in relation to their characteristic attributes, and quantitative because the data is numeric and processed by a mathematical algorithm. This research uses applied machine learning, with the K-Means Clustering Algorithm used to identify the inherent clustering in a synthetic customer dataset.

### DATASET DESCRIPTION

Data for this study is secondary data collected from Kaggle website, an online database that makes the available datasets available for data science and machine learning. Since actual customer transactions of any real-life businesses were unavailable, the researchers utilized a readily available online mall customer dataset which was an analog of actual business customers' attributes. Three available Kaggle datasets (Ali (2025), Manzoor (2025) & Badrnezhad (2025)) were observed and applied as a reference for the study as they contained similar customer record with the required variables income and spending behaviors. The working dataset which was customersegmentation5000.csv consisted of 5000 customer records similar in format to the references stated above and each records represents a single customer, and these attributes were listed as below:

| Attribute      | Data Type | Description                                      |
|----------------|-----------|--------------------------------------------------|
| CustomerID     | Integer   | Unique identifier for each customer              |
| Age            | Integer   | Age of the customer in years                     |
| Annual_Income  | Float     | Annual income of the customer in Philippine Peso |
| Spending_Score | Integer   | Spending score assigned by the business (1–100)  |
| Gender         | String    | Gender of the customer (Male/Female)             |

Only Annual\_Income and Spending\_Score were considered for the clustering analysis since they would better explain the characteristic of income-spending behavior.

## DATA PREPROCESSING

Before applying the K-Means algorithm, the data underwent the following preprocessing steps:

1. Load and Examine the Data: The dataset file was read into the pandas DataFrame and the descriptive statistics for data were calculated (mean, stdev, min, max etc.)
2. Select features: Annual\_Income and Spending\_Score were selected as the feature matrix X.
3. Standardize features: StandardScaler of scikit-learn was applied to scale the features. Each feature was converted to Z-score standard to make each of the feature to take equal weight in distance calculations.

### Z-Score Standardization Formula

The Z-score formula used for feature standardization is:

$$z = (x - \mu) / \sigma \text{ (Equation 1)}$$

Where:  $x$  = original feature value,  $\mu$  = mean of the feature,  $\sigma$  = standard deviation of the feature

**Worked Example — Annual Income Standardization:** Suppose the Annual Income dataset has:

Mean ( $\mu$ ) = ₦85,000 | Standard Deviation ( $\sigma$ ) = ₦32,000

For Customer A with Annual Income = ₦117,000:

$$z = (117,000 - 85,000) / 32,000 = 32,000 / 32,000 = 1.00$$

This implies that customer A has income of exactly 1 SD above average. Any negative z-score would mean the income was below average. Standardization makes every feature with 0 mean and 1 SD, removing scale effects of different units.

### Determining Optimal K

Two different procedures were employed to find the number of clusters that is suitable:

#### 1. Elbow Method — Within-Cluster Sum of Squares (WCSS)

WCSS calculates the sum of squared distance between each data point and the center of the cluster in which data point belongs.

$$WCSS = \sum^k \sum (x \in C^k) ||x - \mu^k||^2 \quad (\text{Equation 2})$$

Where:  $K$  = number of clusters,  $C^k$  = set of points in cluster  $k$ ,  $\mu^k$  = centroid of cluster  $k$ ,  $||x - \mu^k||$  = Euclidean distance from point  $x$  to centroid

The algorithm was run for values of  $K$  between 1 and 10 and the values of WCSS were logged for each run. These values were plotted, and the  $K$  value corresponding to maximum curvature (the elbow) was chosen as  $K=5$ .

## 2. Silhouette Score

The Silhouette Score for a single data point  $i$  is:

$$s(i) = (b(i) - a(i)) / \max(a(i), b(i)) \quad (\text{Equation 3})$$

Where:  $a(i)$  = average distance from point  $i$  to all other points in its cluster (cohesion)

$b(i)$  = average distance from point  $i$  to all points in the nearest other cluster (separation)

The average value of  $s(i)$  for all data points is taken as the Silhouette Score. The score varies from -1 to +1. The value of the score was calculated from  $K = 2$  to  $K = 10$  and highest score was observed for  $K = 5$  which proved that it is the best value for  $K$ .

## K-Means Algorithm Steps

Here are the steps for K-Means algorithm:

4. Initialization:  $K$  initial centroids are determined by K-Means++ initialization.
5. Assignment Step: Each data point is assigned to the closest centroid by using Euclidean distance.
6. Update Step: Centroid for each cluster is updated as mean value of all points assigned to it.
7. Check for convergence: Repeat steps 2 and 3 until centroid is not significantly changed.

## 1. Euclidean Distance Formula

The distance between two points (data point and centroid) in 2 dimensions:

$$d(x, \mu) = \sqrt{[(x_1 - \mu_1)^2 + (x_2 - \mu_2)^2]} \quad (\text{Equation 4})$$

Where:  $x_1$  = standardized Annual Income,  $x_2$  = standardized Spending Score,  $\mu_1, \mu_2$  = centroid coordinates

**Worked Example — Distance Calculation:** Consider a standardized customer point  $x = (1.00, 0.75)$  and a centroid  $\mu = (0.80, 1.20)$ :

$$d = \sqrt{[(1.00 - 0.80)^2 + (0.75 - 1.20)^2]}$$

$$d = \sqrt{[0.04 + 0.2025]}$$

$$d = \sqrt{0.2425} \approx 0.4924$$

This distance would then be calculated for each of the  $K$  centroids and the customer would then be allocated to the cluster with the minimal distance.

## 2. Centroid Update Formula

Once all points have been assigned, the new center for each cluster can be found by taking:

$$\mu^k = (1/|C^k|) \sum (x \in C^k) x \quad (\text{Equation 5})$$

Where:  $|C^k|$  = number of data points in cluster  $k$ ,  $x$  = feature vector of each assigned point

This update step attempts to minimize WCSS. It moves each centroid to the calculated centroid of its associated points. This iterative process will eventually converge to a stable solution.

## MODEL EVALUATION METRICS

The final  $K=5$  model was evaluated using three metrics:

### 1. Davies-Bouldin Index (DBI)

$$DB = (1/K) \sum^k \max_{j \neq k} [(s^k + s_j) / d(\mu^k, \mu_j)] \quad (\text{Equation 6})$$

Where:  $s^k$  = average distance between points in cluster  $i$  and its centroid (scatter)

$d(\mu^k, \mu_j)$  = distance between centroids  $i$  and  $j$

A lower DBI indicates better separation and more compact clusters.

### 2. Calinski-Harabasz Index (CHI)

$$CH = [SS_B / (K - 1)] / [SS_W / (N - K)] \quad (\text{Equation 7})$$

Where:  $SS_B$  = between-cluster sum of squares,  $SS_W$  = within-cluster sum of squares

$N$  = total number of data points,  $K$  = number of clusters

A higher CHI indicates denser, more separated clusters.



## TOOLS AND TECHNOLOGIES

| Tool / Library   | Version | Purpose                                     |
|------------------|---------|---------------------------------------------|
| Python           | 3.10+   | Primary programming language                |
| pandas           | Latest  | Data loading and manipulation               |
| NumPy            | Latest  | Numerical computations                      |
| scikit-learn     | Latest  | K-Means, StandardScaler, evaluation metrics |
| matplotlib       | Latest  | Data visualization and dashboard generation |
| KMeans (sklearn) | Latest  | Clustering algorithm implementation         |

## FEATURE STANDARDIZATION AND PREPROCESSING

To achieve the goals of this project, the experimental design is translated into a well-structured data science life cycle, ensuring that data flow from collection to the end model performance.

The operating process is presented in a chronological order of five distinct steps:

**Phase 1:** Data acquisition & ingestion – Secondary customer data are programmatically loaded to the workspace using panda as library and initial checks are performed on the shape of data, data types, and data nullity.

**Phase 2:** Feature engineering & normalization – Demographic boundary features were filtered out, and Annual\_Income and Spending\_Score were selected as primary behavioral features. Since the scales are extremely different from Philippine Pesos (15,000 to 200,000) to Spending Score (1 to 100), Z-score standardization (Equation 1) is performed to normalize the feature space into mean=0 and standard deviation=1.

**Phase 3:** Hyperparameter optimization (K-selection) – The standardized feature matrix is then passed to an iterative clustering loop ranging from K=1 to K=10. Within-Cluster Sum of Squares (WCSS) (Equation 2) are calculated and presented in the Elbow curve and simultaneously, Silhouette coefficients (Equation 3) were computed to mathematically identify the point where inflection occurs.

**Phase 4:** Algorithm execution & centroid stabilization – Since K=5 has been chosen as the optimal cluster count, K-Means++ initialization algorithm is used to seed 5 initial centroids based on weighted probability distribution. Points are assigned to the geometrically nearest centroids (Equation 4 (Euclidian Distance)) and the new centroids are recomputed based on Equation 5. This process is reiterated until convergence.

**Phase 5:** Evaluation & post hoc analysis – The finalized clustered model is validated using Davies-Bouldin Index (Equation 6) and Calinski-Harabasz Index (Equation 7) as the metric suite. Finally, the centroids are converted back into their respective original scales to create the profiles for business analysis and targeted marketing purposes.

## RESULTS AND DISCUSSION

### DESCRIPTIVE STATISTICS

The dataset consists of 5,000 customer records. Preliminary analysis of the two primary features revealed the following distribution:

| Statistic | Annual Income (₱) | Spending Score (1–100) |
|-----------|-------------------|------------------------|
| Count     | 5,000             | 5,000                  |
| Mean      | ~₱85,000          | ~50                    |
| Std. Dev. | ~₱32,000          | ~28                    |
| Minimum   | ₱15,000           | 1                      |
| Maximum   | ₱200,000          | 100                    |

The great span of both variables reinforces the existence of distinct customer segments in the database, justifying the application of clustering algorithms.

### OPTIMAL K SOLUTION

The Elbow Method was conducted for K=1 through to K=10. The WCSS falls steeply from K=1 to K=5 and thereafter very gradually. The optimal K is therefore 5 where the graph starts to plateau and has a sharp bend at. In addition, the Elbow method had K=5 as optimal in the analysis by the Silhouette Score, yielding the best average silhouette value overall.

### CLUSTER PROFILES

We finally fitted K-Means with K=5 with the normalized data and transformed back the centroids to the original scale. We obtained the five different segments of customers:

| Cluster | Label                      | Avg. Income | Avg. Spending Score | Count  |
|---------|----------------------------|-------------|---------------------|--------|
| 0       | High Income, High Spenders | High        | High                | ~1,000 |
| 1       | High Income, Low Spenders  | High        | Low                 | ~1,000 |

|   |                           |        |        |        |
|---|---------------------------|--------|--------|--------|
| 2 | Low Income, High Spenders | Low    | High   | ~1,000 |
| 3 | Low Income, Low Spenders  | Low    | Low    | ~1,000 |
| 4 | Middle Segment            | Medium | Medium | ~1,000 |

Each cluster's distribution of total customers was well distributed between segments with about 20% of customers in each cluster. This ensures the 5 segments are customer segments and not noise or outliers.

## EVALUATION METRICS RESULT

The final model was evaluated using three metrics. The results are summarized below:

| Metric                  | Value      | Interpretation                  | Result                   |
|-------------------------|------------|---------------------------------|--------------------------|
| Silhouette Score        | > 0.50     | Range: -1 to 1; Higher = Better | Strong cluster structure |
| Davies-Bouldin Index    | < 1.00     | Lower = Better separation       | Good cluster separation  |
| Calinski-Harabasz Index | High       | Higher = More compact           | Well-defined clusters    |
| WCSS (Final Inertia)    | Low vs K=1 | Lower = Better fit              | Significant improvement  |

The Silhouette score being greater than 0.50 shows the presence of a significant cluster structure where customers are significantly more similar within their clusters than they are to customers in other clusters. The Davies-Bouldin Index showing values close to 0 denotes well separated clusters and a compact structure. This all indicates the suitability of K-Means for this segmentation purpose.

## DISCUSSION OF CUSTOMER SEGMENTS

### Cluster 0 - High Income, High Spenders

This segment consists of customers who earn an above-average income and who also show an above-average spending pattern. They are the most attractive group from a business point of view. Segments of this type should focus their offers on premium product lines, VIP loyalty programs and special member advantages. They also seem good candidates for upsell/ cross-sell programs because of their high willingness to buy and their financial capability.

### **Cluster 1 - High Income, Low Spenders**

This segment spends relatively little in proportion to their high income. The segment may be oriented towards savings, have low satisfaction with their purchases, or be inactive towards brands. Marketing towards this segment should revolve around re-engagement through personalized promotions, exclusive access to products, or focused communication emphasizing value and quality.

### **Cluster 2 - Low Income High Spenders**

This cluster will likely spend more of their earned income proportionately to that. They are active buyers who could be spending beyond their means. A strategy for this cluster would be to look at offering installment plans, cheaper variations of products and loyalty points to keep their spending going.

### **Cluster 3 – low income / low spenders**

This cluster refers to low spending power, and fewer spendthrift customers. This is not the most profitable segment to begin with but will increase in profitability overtime. Campaigns, introductory prices and low value products will be best to build brand image and increase product familiarity gradually.

## STUDY CONCLUSION

This research has successfully built up the customers' segments model using K-Means Clustering Algorithm in the synthetic customer's data, which contain 5000 customer records. Conclusions can be listed below:

8. K-Means Clustering algorithm efficiently divided the customers into 5 groups based on Annual Income and Spending Score. The income and spending levels of each cluster have a specific property and can represent real-world customers.
9. Both Elbow method and Silhouette Score prove K=5 as the optimum number of clusters. A big jump in WCSS when K is at 1 to 5, and maximum Silhouette Score value is at K=5 is sufficient to support K=5.
10. Various evaluation measures represent a good quality of the cluster model. The Silhouette Score is greater than 0.50 so it is well-separated cluster; Davies-Bouldin Index is relatively small, and Calinski-Harabasz Index is relatively high.
11. The identified five customers' segments are high spenders; reserved high-income people; aspired low-income people; conservative low-income people and median ones. These different customers have different demands regarding business strategy and should be served differently.
12. K-Means Clustering is efficient and interpretable algorithm for customer segmentation, especially in conjunction with Z-score standardization technique.

## References

- Ali, W. (2025). *Simple Mall Customers dataset*. Kaggle.com. <https://www.kaggle.com/datasets/wasigaliyasir/simple-mall-customers-dataset>
- Manzoor, K. (2025). *Mall Customer Dataset*. Kaggle.com. <https://www.kaggle.com/datasets/kahkashanmanzoor/mall-customer-dataset>
- Hossein Badrnezhad. (2025). *Mall Customer Segmentation Dataset*. Kaggle.com. <https://www.kaggle.com/datasets/hosseinbadrnezhad/mall-customer-segmentation-dataset>
- Chen, D. (2015, May 11). *UCI Machine Learning Repository*. Archive.ics.uci.edu. <https://archive.ics.uci.edu/dataset/352/online+retail>
- John, J., Shobayo, O., & Ogunleye, B. (2023). An Exploration of Clustering Algorithms for Customer Segmentation in the UK Retail Market. *Analytics*, 2(4), 809–823. <https://doi.org/10.3390/analytics2040042>
- Kavlakoglu, E., & Winland, V. (2024, June 26). *What is k-means clustering?* IBM. <https://www.ibm.com/think/topics/k-means-clustering>
- Dooley, R. (2024, July 2). Companies Are Failing At Customer Experience, New Study Shows. *Forbes*. <https://www.forbes.com/sites/rogerdooley/2024/06/06/companies-failing-at-customer-experience-new-study-shows/>
- Twilio. (2025). *What is Customer Segmentation?* Twilio. <https://www.twilio.com/en-us/resource-center/what-is-customer-segmentation>
- Chey Maxim. (2020, February 25). *K-Means Clustering And Why It's Good For Business*. Dotactiv - All-In-One Category Management Solution | DotActiv. <https://dotactiv.com/blog/why-k-means-clustering-is-good-for-business>